

Cadence

Daily tasks. Real life. One calm surface.

Product Specification — v1.0

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Working name: Cadence (final name TBD — alternates listed in Appendix A)

Status: Draft for pre-build review

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1. Executive Summary

Cadence is a polished, calm task and calendar companion for people who want one app to run their day across personal life and business — without feeling like enterprise software.

It does five things deliberately well: (1) a single Today view that mixes tasks and calendar events, (2) automatic rollover so incomplete work never falls off the radar, (3) shared lists for joint business with a co-owner, (4) deep iOS integration via widgets, Lock Screen, Siri, and notifications, and (5) an AI layer powered by Claude that helps plan the day and capture work in natural language.

The product is designed iPhone-first and Mac-second, using a single SwiftUI codebase. The MVP can be in your hands within roughly 6–8 focused weeks of build time; v2 (Siri + Claude) adds another 3–4 weeks.

Why this exists

You evaluated TickTick, Things 3, Apple Reminders, Notion, Todoist, and others. Each fails at least one core need:

- **TickTick** — feature-rich but visually cluttered; weak daily dashboard; hard to see what matters.
- **Things 3** — beautiful but no shared lists with non-Things users; weak Google Calendar; no Claude.
- **Reminders** — too basic; no real list-sharing model that scales; no rich widgets.
- **Todoist** — good shared lists but uninspired UI and walled-off premium features.
- **Notion** — too heavy; no glanceable iPhone experience; offline is fragile.

Cadence is opinionated about being a daily-driver, not a project management system. It is lean by design.

2. Vision & Principles

Vision

A daily-driver task and calendar app that is beautiful enough to enjoy opening every morning, intelligent enough to keep up with messy real life, and connected enough to be the single source of truth across all the contexts of your life.

Design Principles

- **Glance before grind.** The first thing you see is curated — today's tasks and events, nothing more. Power lives one tap deeper.
- **Rollover by default.** Software should adapt to humans, not the other way around. Incomplete work moves forward on its own.
- **Lists are contexts.** Personal, Business, Joint Business, Side Project — each has its own color, icon, and sharing model.
- **Calendar as a peer.** Tasks and events live in the same surface. Your day is one timeline.
- **One model, every surface.** iPhone, Mac, widgets, Lock Screen, Siri — same data, same interactions, no jarring shifts.
- **AI as a teammate.** Claude reads your day, drafts plans, and turns rambling thoughts into clean tasks — but never acts without permission.
- **Quietly beautiful.** Spring animations, refined typography, generous whitespace, deliberate haptics. Polish is a feature, not a finish.

3. Target User & Use Cases

Primary user

You — Tripp. A multi-hat operator: business co-owner plus personal life. iPhone-first. Mac at the desk. Values aesthetics and dislikes software that fights you. Wants Siri and Claude to be real, not gimmicks.

Secondary user

Your business co-owner. They collaborate via shared Business lists. They may or may not have an iPhone — so the data model and (eventually) the web view need to accommodate non-Apple devices.

Top use cases

- **Morning brief.** Open phone → glance at widget → see 4 tasks + 2 meetings → tap one to start.
- **Mid-day capture.** Driving, get an idea → 'Hey Siri, add to Cadence: call insurance broker tomorrow.'
- **Shared business task.** Add 'review Q2 numbers' to Joint Business list → co-owner sees it instantly with a notification.
- **End-of-day reflection.** See what rolled over → snooze, reschedule, or knock out the last few.
- **Weekly planning (v2).** 'Plan my week' button → Claude reads calendar + open tasks → proposes time blocks → you accept or edit.

4. Feature Scope by Version

MVP is everything required for Cadence to replace TickTick on your phone. v1.1 expands to Mac and adds power-user features. v2 layers in the AI and Siri experience.

Feature	MVP v1.0	v1.1	v2 (AI)	Notes
Tasks (CRUD, due, priority, subtasks, tags)	✓	—	—	Core. Edit/complete in ≤ 2 taps.
Lists (Personal, Business, custom)	✓	—	—	Unlimited lists. Color + icon per list.
Auto-rollover of incomplete tasks	✓	—	—	Default on. Per-list override.
Today view (tasks + events)	✓	—	—	First screen on launch.
Google Calendar (read-only)	✓	—	—	Multi-calendar selection.
Sharing (per-list collaborator)	✓	—	—	Invite by email. Edit / view roles.
Push notifications + daily brief	✓	—	—	Configurable per-list and per-task.
iOS Home Screen widgets (S/M/L)	✓	—	—	Today snapshot. Tap to deep-link.
Lock Screen widgets (rect + circular)	✓	—	—	iOS 16+ accessory family.
iCloud sync (iPhone ↔ Mac)	✓	—	—	CloudKit private DB.
Native Mac app (SwiftUI)	—	✓	—	Same codebase via Apple multi-platform.
Recurring tasks (RFC 5545 rules)	—	✓	—	Daily / weekly / monthly / custom.
Smart filters (this week, no date, etc.)	—	✓	—	User-creatable saved filters.
Apple Watch complication	—	✓	—	Glanceable count + next task.

Feature	MVP v1.0	v1.1	v2 (AI)	Notes
Siri Shortcuts (add / list / complete)	—	—	✓	App Intents framework.
Claude integration (plan day, NL entry)	—	—	✓	Anthropic API. User brings own key or subscribes.
Two-way Google Calendar sync	—	—	✓	Create events from time-blocked tasks.
Time tracking + weekly review	—	—	✓	Light habit-style retrospectives.

5. MVP Feature Specs

5.1 Tasks

Properties

- Title (required)
- Notes (long text, markdown supported)
- Due date (optional) and Due time (optional)
- List (required, defaults to Inbox)
- Priority: None / Low / Medium / High
- Tags (free-form, used for filtering)
- Subtasks (one level deep, MVP)
- Reminder offsets (e.g., at due time, 30 min before — multiple allowed)
- Created by, Completed by (for shared lists)

Interactions

- **Add.** Tap '+' anywhere → inline composer slides up. Natural language parsed: 'Email Sam tomorrow at 3pm #business' auto-fills date, time, list.
- **Complete.** Tap circle → spring + haptic + strike-through, task fades into completed group after 2 seconds (with undo).
- **Edit.** Tap task body → detail sheet. Swipe right on row → quick complete; swipe left → snooze / delete.
- **Drag.** Long-press → drag to reorder within list or move between lists.

Edge cases

- Deleting a task with subtasks confirms — 'Delete 3 subtasks too?'
- Completing a parent does NOT auto-complete subtasks (user choice).
- Tasks without a due date never roll over — they live in their list until completed.

5.2 Lists

Defaults

- Inbox (catch-all, can't be deleted)
- Personal
- Business (private to you)

- Joint Business (shared with co-owner)

Properties

- Name, Color (12 preset palette), SF Symbol icon
- Visibility: Show in Today / Hidden
- Sharing: Private / Shared (with collaborators list)
- Per-list defaults: rollover on/off, default reminder offset, default priority

Behavior

- Lists can be reordered. Hidden lists still receive tasks but don't surface in Today.
- Deleting a list with tasks asks: move tasks to Inbox, or delete with the list.

5.3 Today View (the heart of the app)

Layout

- Top: friendly date header — 'Wednesday, May 13'
- Stat strip: tasks due, events today, carried over count
- Carried-over section (collapsible): yesterday's incomplete tasks with subtle 'carried over' chip
- Timeline section: tasks with times + calendar events, interleaved chronologically
- Unscheduled section: tasks due today with no time
- Quick-add bar pinned at bottom; tap to expand to NL composer

Behavior

- Pull to refresh syncs calendar + CloudKit
- Pinch out to see week view (events + task heatmap)
- Calendar events are read-only in MVP; tap opens a sheet with attendees, location (with map link), notes

5.4 Auto-Rollover (the differentiator)

Default behavior

- Any task with a due date < today and status = incomplete is treated as 'carried over' today.
- Carried-over tasks appear in Today view in a dedicated section at the top, with a chip showing original due date ('was Mon').
- Completing a carried-over task closes it normally; its history records both original due date and completion date.

Per-task / per-list overrides

- **Rollover: On (default)** — behavior above.
- **Rollover: Ask each morning** — first launch of the day shows a sheet: 'You have 4 carried-over tasks. Bring forward? Reschedule? Snooze?'
- **Rollover: Off** — task simply shows as 'Overdue' with a red chip. Doesn't move.

Snooze

- Swipe → Snooze → quick chips (Tonight, Tomorrow, This Weekend, Next Week, Pick a date)
- Snoozed tasks hide until snooze date, then reappear in Today

Edge cases

- Recurring tasks don't roll over — completing/skipping the current instance is handled by the recurrence rule (v1.1).
- Subtasks roll with their parent.
- Tasks rolled over while user was on vacation: 'You have 14 carried-over tasks — review them?' modal on first open if count > 10.

5.5 Sharing

Sharing model

- Sharing is per-list, not per-task.
- Owner can invite by email; invitee receives a deep link and accepts in-app.
- Roles: Owner (full), Editor (add/edit/complete), Viewer (read only).
- Owner can transfer ownership or remove collaborators.

Visibility & attribution

- On shared lists, each task displays a tiny avatar of who created it; completing shows who completed it.
- Activity feed per shared list: 'Co-owner completed Review Q2 numbers — 9:42 AM.'
- Notifications fire on shared-list changes per user preferences (default: notify on @mentions only).

Conflict resolution

- Last-write-wins per field, with CloudKit conflict surfacing for non-trivial conflicts (e.g., two people edit the title within 5 seconds).
- Offline edits queue and reconcile on reconnect.

5.6 Google Calendar Integration (read-only, MVP)

Setup

- Settings → Connected Accounts → Sign in with Google (OAuth 2.0 via ASWebAuthenticationSession).
- Pick which calendars to display; each calendar gets a color override.
- Token refresh handled silently; user notified if reauth required.

Behavior

- Events fetched in a sliding window: today – 7 days to today + 30 days.
- Events appear in Today view interleaved with tasks (sorted by start time).
- Tap event → detail sheet (read only). Includes 'Open in Google Calendar' deep link.
- All-day events appear in the date header strip, not in the timeline.
- Background refresh every 15 min when app is foreground; silent push refresh in background (v2).

Privacy

- OAuth tokens stored in Keychain. Calendar payloads never leave the device.
- Display-only — calendar data is not written to CloudKit, only cached locally.

5.7 Notifications

Types

- **Task reminders.** Per-task, multiple offsets (at time, 5 min before, 1 hr before, day before).
- **Morning brief.** Once per day at user-set time (default 7:30 AM). 'You have 5 tasks and 2 meetings today.'
- **Event reminders.** Mirror Google Calendar event reminders, with user-configurable suppression per calendar.
- **Sharing notifications.** Shared list changes — opt-in granularity (everything / @mentions only / off).

Implementation

- UNNotificationCenter for local; APNs for shared-list pushes.
- Notifications grouped by list with a Cadence summary.
- Critical Alerts not used (no entitlement needed).
- Live Activity (iOS 16+) for the next event/task: optional, off by default.

5.8 Widgets & Lock Screen

Home Screen widgets

- **Small.** Count of tasks today + next task title.
- **Medium.** Today list — up to 4 tasks, tappable to deep-link.
- **Large.** Today list (up to 6) + next 3 events with times.

Lock Screen accessories (iOS 16+)

- **Rectangular.** Next task or event with countdown.
- **Circular.** Tasks-due count.
- **Inline.** '4 tasks · next: 2:30 PM Sam call'

Behavior

- Widgets refresh on a 15-minute timeline + reload on task completion.
- Widgets respect interactive widgets API (iOS 17+) — tap to complete without opening app.
- Use WidgetKit + AppIntents for interactivity.

5.9 Sync (iPhone ↔ Mac)

- Storage: SwiftData (Core Data + Codable) backed by CloudKit Private Database.
- Shared lists use CloudKit Shared Database; CKShare per shared list.
- Conflict policy: last-write-wins per field; CloudKit handles offline merge.
- All client devices subscribe to CKDatabaseSubscriptions for silent push refresh.
- Targets: <1s sync latency on Wi-Fi between two devices.

5.10 Onboarding

- 3 screens: Welcome → 'Connect your calendar' → 'Pick your lists.'
- Import from TickTick: paste exported CSV, mapped to lists. v1.0 supports CSV; v1.1 adds native importers.
- Notification permission requested after first task is added, not upfront.
- Onboarding total time target: < 60 seconds.

6. v1.1 — Mac, Power Tools, Watch

- Native Mac app — same SwiftUI codebase via multi-platform target. Keyboard-first interactions. Menubar 'mini' app.
- Apple Watch complication: tasks-due count, next event.
- Recurring tasks with RFC 5545 RRULEs.
- Smart filters: 'This week,' 'No due date,' 'Overdue > 3 days,' user-creatable.
- Bulk actions: multi-select + reschedule/move/complete.
- Markdown rendering in task notes.
- Apple Reminders import.

7. v2 — Siri + Claude

Siri Shortcuts (App Intents)

- "Add to Cadence" — natural language captured as a new task. Disambiguates list if ambiguous.
- "What's on my plate today" — Siri reads today's tasks and event count.
- "Mark X done" — fuzzy match on task titles in Today.
- "Start my day" — opens Cadence to Today view (Shortcut + Personal Automation friendly).

Claude integration

Plan my day

- Big button in Today header — 'Plan my day with Claude.'
- Sends: today's tasks (titles, due times, priority), calendar events, work hours. Receives: a proposed time-block schedule and a written summary.
- Output rendered as draft blocks; user taps Accept to convert to time-blocked tasks.

Capture

- Long-press the '+' button → voice input → Claude parses into one or more tasks with structured fields. Confirm-or-edit before save.

Weekly review

- Friday afternoon prompt — 'Want a Claude-powered review?' Claude summarizes the week (completed, carried-over patterns, calendar pressure) and suggests focus areas for next week.

Privacy & cost

- Default: app uses its own Anthropic API key (subsidized by an optional Cadence Plus subscription). Power users can supply their own key for raw cost.
- Only the user's task titles + calendar event titles are sent — never notes by default; notes are opt-in per task.
- All AI exchanges logged locally (viewable in Settings → AI History) and never used for training.

8. v3 — Two-way Sync, Time Tracking, Future

- Two-way Google Calendar sync: time-blocked tasks become calendar events.
- Apple Calendar two-way sync (EventKit).
- Time tracking: start/stop timers per task; report totals per list per week.
- Habits: light-weight recurring 'streak' tasks with a habit view.
- Web companion for the co-owner (read/edit shared lists from any browser).
- Android version (only if co-owner adoption requires it).

9. Non-Goals (what we are NOT building)

Saying no is half the design. Cadence is deliberately not:

- A project management tool (no kanban, no Gantt, no dependencies between tasks).
- A note-taking app (notes are simple; rich docs belong in Notes/Notion).
- An enterprise team collaboration platform (designed for 1:1 sharing; not Asana).
- A habit tracker (until v3, and only lightly).
- A finance/budget tool, email client, or browser.
- A platform with public sharing or community features.

10. Success Criteria

MVP launch criteria

- Cold-launch to Today view in under 1 second on iPhone 14+.
- Complete a task in ≤ 2 taps from any state (including widget).
- Onboarding completion in < 60 seconds for a new user.
- TickTick CSV import succeeds for a 500-task export.
- Two devices stay in sync within 2 seconds on Wi-Fi.
- Crash-free session rate $\geq 99.5\%$.
- Accessibility: VoiceOver fully usable; Dynamic Type up to XXXL.

Long-term success

- You stop opening TickTick / Reminders / Notes for daily tasks.
- Co-owner uses Joint Business list regularly without prompting.
- AI 'Plan my day' is used ≥ 3 days/week within 30 days of v2 launch.

11. Open Questions to Resolve Before Build

- **Final name.** 'Cadence' is the working name. Appendix A lists alternates.
- **Distribution.** Personal app on TestFlight forever (free), or eventual App Store launch? Affects pricing model and legal entity.
- **Pricing model.** If shipped publicly: free MVP, Cadence Plus (\$3.99/mo or \$29.99/yr) for AI features and sharing? Free for you regardless.
- **Co-owner platform.** Confirm co-owner's device (iPhone? Android?) — drives whether web companion is needed in v1.1 vs v2.
- **Calendar provider scope.** Google Calendar in MVP. Apple Calendar in v1.1. Outlook/Microsoft 365 — desired?
- **Anthropic API approach.** Subsidized via subscription, or BYOK only? Affects cost model and v2 timeline.

Appendix A — Name Candidates

All are .com-checkable and short. None are guaranteed available — final pick should run a trademark + domain check.

- **Cadence (working name).** Rhythm, daily flow. Strong, mature, memorable.
- **Tide.** Tasks roll forward like a tide. Calm, soft, easy to remember.
- **Loop.** Closed loop on tasks. Modern, short, has tech-startup energy.
- **Daycard.** A card for each day. Concrete, friendly.
- **North.** Daily compass. Aspirational.
- **Tend.** You tend to your tasks. Warm, slightly novel.
- **Drift.** Tasks drift forward — but might read as 'lazy.' Probably skip.

Appendix B — Glossary

- **Carried over.** Internal status for a task whose due date < today and is still incomplete.
- **Joint Business.** Default name for the shared business list between you and your co-owner.
- **List.** A named container for tasks. Has color, icon, sharing settings.
- **Today view.** The default screen on launch — tasks + events for the current day.
- **Snooze.** Push a task's due date forward by a chosen amount.
- **Rollover.** Automatic carry-forward of incomplete tasks to today.
- **Inbox.** Default destination list for tasks created without an explicit list.
- **Plan my day.** v2 Claude action — proposes a time-blocked plan from today's tasks + events.